

- 8 A proposed space laboratory is to create artificial gravity as shown in Fig. 8.a and Fig. 8.b below.

The space laboratory is rotated about an axis so that it simulates an acceleration due to gravity equal to that of the gravity due to Earth at the outer ring which has a radius R_1 of 2150 m.

What should be the approximate radius R_2 of the inner ring, so that it simulates the acceleration due to the gravity of Mars which is 3.72 m s^{-2} ?

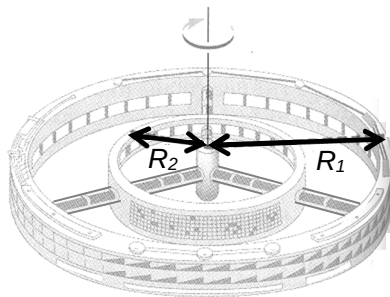


Fig. 8.a

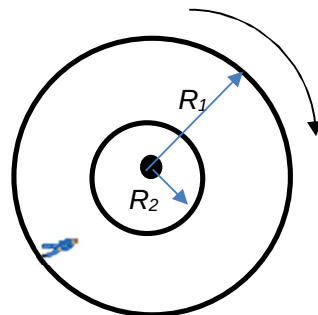


Fig. 8.b

A 700 m

B 800 m

C 900 m

D 1000 m

[Turn over

- 9 A person is located near the city of Shanghai which has a latitude of 31.28° N